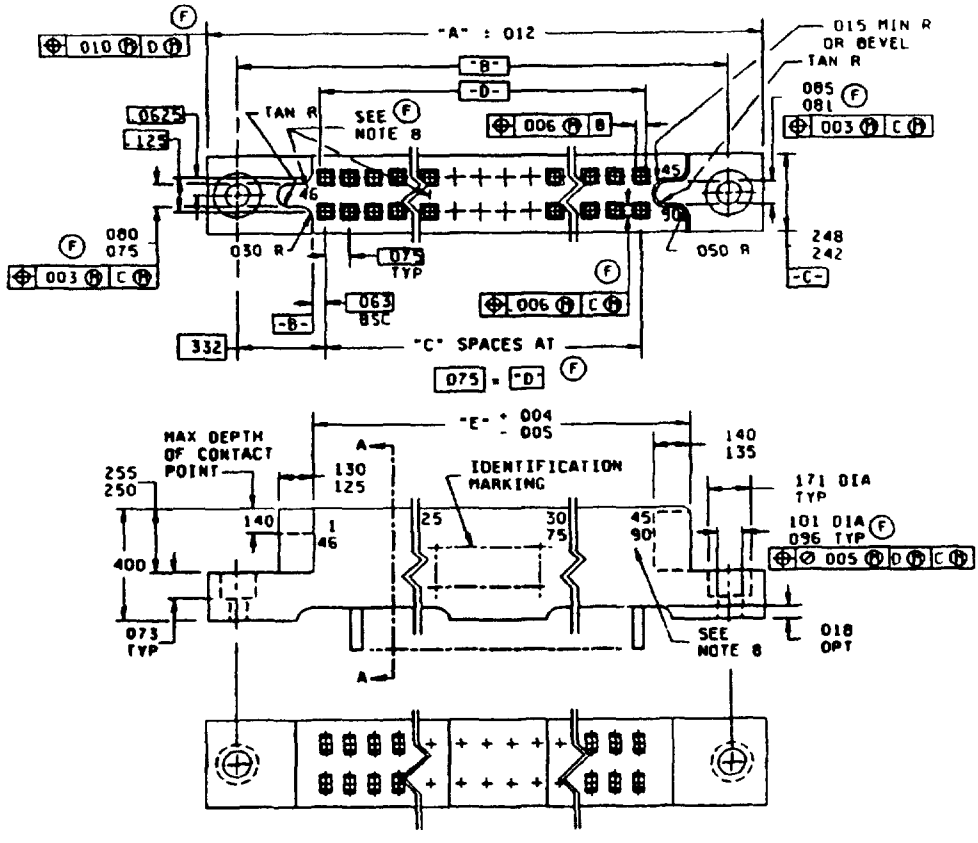
INCH-POUND
MIL-C-55302/24F
7 June 1993
SUPERSEDING
MIL-C-55302/24E
31 October 1988

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLE, DECADE INCREMENT 10 THRU 110 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.075 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55302



STYLE 1 (INTERNAL KEYING)

(F) FIGURE 1 Connectors, receptacle, 075 (1.90 mm) spacing

(F) denotes changes

DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

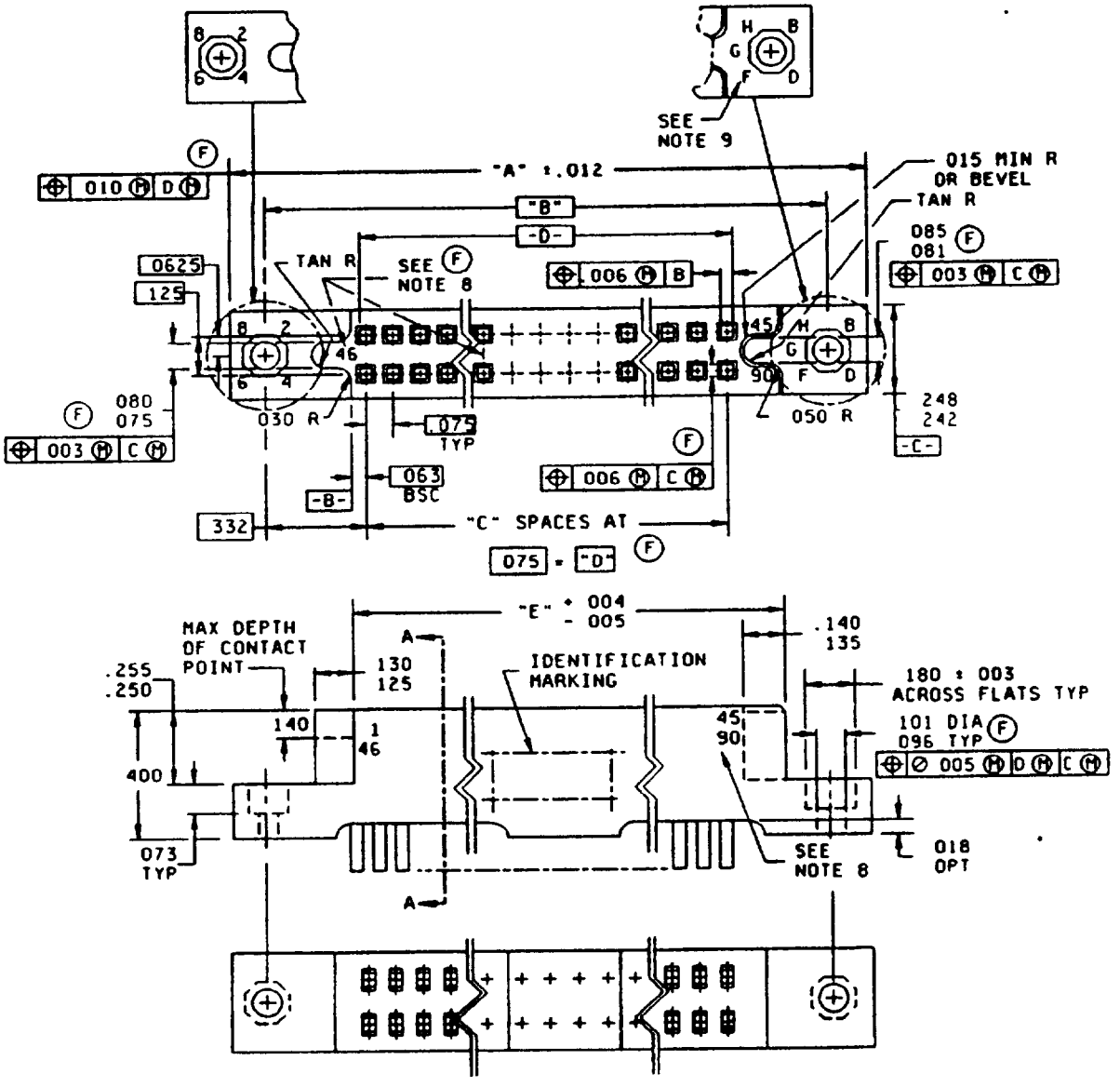
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Dash 1/ numbers	Number of contacts	Type	Dimensions					Contact identification numbers
			A	B	C	D	E	
01 02 03 04	10	I II III IV	1.255 (31 88)	965 (24 51)	4	.300 (7 62)	.565 (14.35)	1 - 5 6 - 10
05 06 07 08	20	I II III IV	1 630 (41 40)	1 340 (34 04)	9	.675 (17 15)	940 (23 88)	1 - 10 11 - 20
09 10 11 12	30	I II III IV	2 005 (50 93)	1 715 (43 56)	14	1 050 (26 67)	1 315 (33 40)	1 - 15 16 - 30
13 14 15 16	40	I II III IV	2 380 (60 45)	2 090 (53 09)	19	1 425 (36 20)	1 690 (42 93)	1 - 20 21 - 40
17 18 19 20	50	I II III IV	2 755 (69 98)	2.465 (62 61)	24	1.800 (45.72)	2.065 (52 45)	1 - 25 26 - 50
21 22 23 24	60	I II III IV	3 130 (79.50)	2.840 (72 14)	29	2.175 (55 25)	2 440 (61 98)	1 - 30 31 - 60
25 26 27 28	70	I II III IV	3 505 (89 03)	3.215 (81.66)	34	2 550 (64 77)	2.815 (71 50)	1 - 35 36 - 70
29 30 31 32	80	I II III IV	3 880 (98.55)	3 590 (91 19)	39	2.925 (74.30)	3.190 (81.03)	1 - 40 41 - 80
33 34 35 36	90	I II III IV	4.255 (108 08)	3 965 (100 71)	44	3 300 (83.82)	3.565 (90 55)	1 - 45 46 - 90
37 38 39 40	100	I II III IV	4 630 (117 60)	4.340 (110 24)	49	3 675 (93 35)	3 940 (100 08)	1 - 50 51 - 100
41 42 43 44	110	I II III IV	5 005 (127 13)	4 715 (119 76)	54	4 050 (102.87)	4 315 (109 60)	1 - 55 56 - 110

1/ Dash numbers 01 through 44 are for style 1 (internal keying) connectors

(F) FIGURE 1 Connectors, receptacle, 075 (1 90 mm) spacing - Continued

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STYLE 2 (EXTERNAL KEYING)

(F) FIGURE 1 Connectors, receptacle, .075 (1.90 mm) spacing - Continued

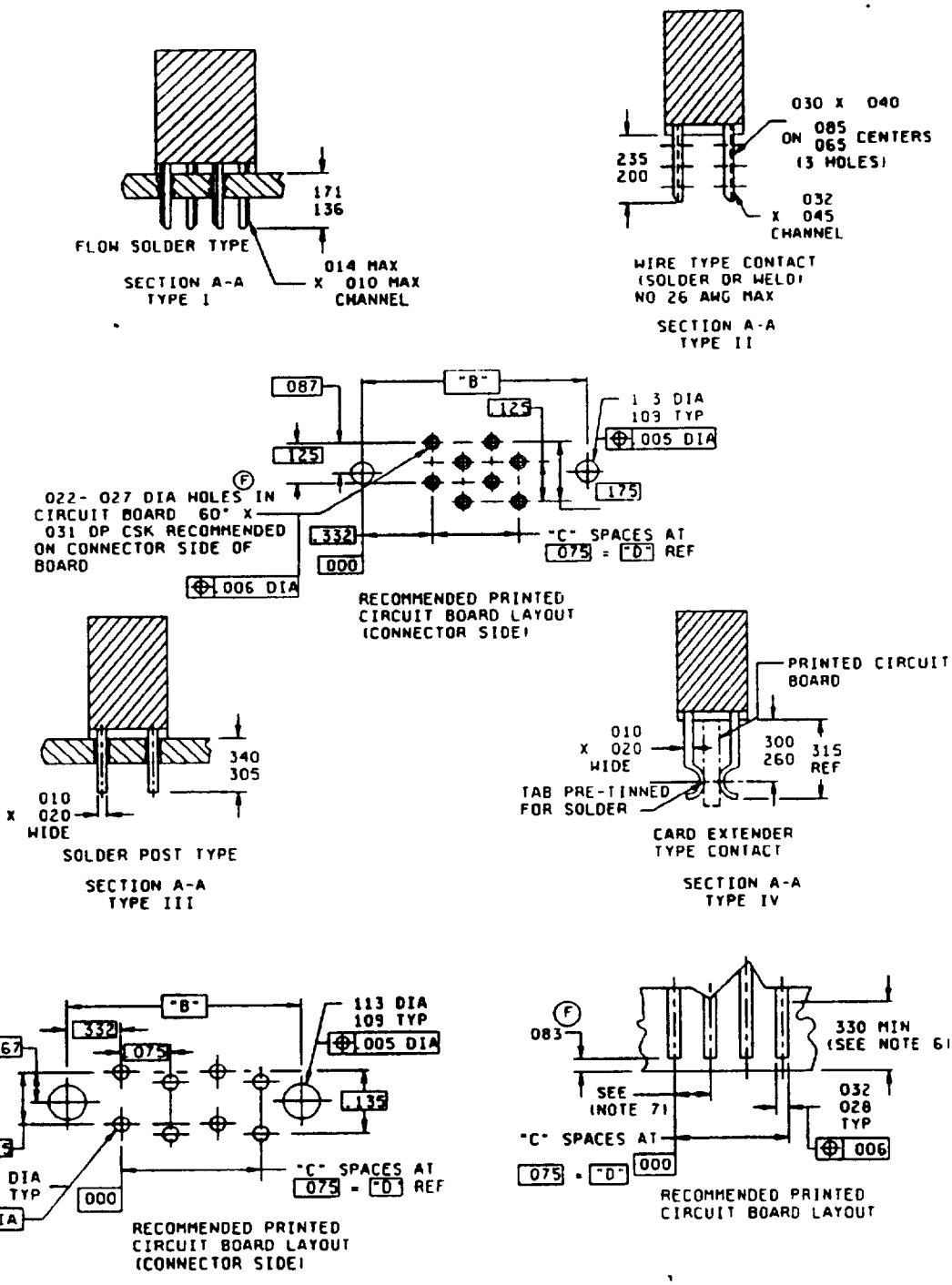
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Dash 1/ numbers	Number of contacts	Type	Dimensions					Contact identification numbers
			A	B	C	D	E	
45 46 47 48	10	I II III IV	1 255 (31 88)	965 (24.51)	4	300 (7 62)	565 (14 35)	1 - 5 6 - 10
49 50 51 52	20	I II III IV	1 630 (41 40)	1.340 (34 04)	9	675 (17 15)	940 (23 88)	1 - 10 11 - 20
53 54 55 56	30	I II III IV	2.005 (50.93)	1.715 (43.56)	14	1 050 (26 67)	1 315 (33.40)	1 - 15 16 - 30
57 58 59 60	40	I II III IV	2 380 (60 45)	2 090 (53 09)	19	1 425 (36 20)	1 690 (42 93)	1 - 20 21 - 40
61 62 63 64	50	I II III IV	2 755 (69 98)	2 465 (62 61)	24	1 800 (45 72)	2 065 (52.45)	1 - 25 26 - 50
65 66 67 68	60	I II III IV	3 130 (79 50)	2 840 (72.14)	29	2.175 (55.25)	2 440 (61 98)	1 - 30 31 - 60
69 70 71 72	70	I II III IV	3 505 (89.03)	3 215 (81 66)	34	2 550 (64 77)	2 815 (71 50)	1 - 35 36 - 70
73 74 75 76	80	I II III IV	3 880 (98 55)	3 590 (91 19)	39	2 925 (74 30)	3.190 (81 03)	1 - 40 41 - 80
77 78 79 80	90	I II III IV	4 255 (108 08)	3.965 (100 71)	44	3 300 (83 82)	3 565 (90 55)	1 - 45 46 - 90
81 82 83 84	100	I II III IV	4.630 (117 60)	4.340 (110.24)	49	3 675 (93 35)	3 940 (100 08)	1 - 50 51 - 100
85 86 87 88	110	I II III IV	5 005 (127 13)	4.715 (119 76)	54	4 050 (102 87)	4 315 (109 60)	1 - 55 56 - 110

1/ Dash numbers 45 through 88 are for style 2 (external keying) connectors

(F) FIGURE 1 Connectors, receptacle, 075 (1 90 mm) spacing - Continued

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F FIGURE 1. Connectors, receptacle, .075 (1.90 mm) spacing - Continued

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Inches	mm	Inches	mm	Inches	mm
.003	0 08	.050	1.27	136	3 45
.004	0 10	.0625	1 588	140	3 56
005	0 13	.063	1 60	.171	4.34
006	0 15	.065	1.65	.175	4.44
.010	0 25	.067	1 70	.180	4.57
012	0 30	.073	1.85	.200	5 08
014	0 36	.075	1 90	235	5.97
015	0.38	.080	2.03	242	6 15
018	0 46	.081	2.06	248	6 30
019	0 48	083	2 11	250	6 35
020	0 51	.085	2 16	255	6 48
022	0 56	087	2 21	260	6 60
025	0 64	096	2 44	300	7 62
027	0 69	101	2 57	305	7 75
028	0 71	109	2.77	315	8 00
030	0 76	113	2 87	330	8 38
031	0 79	125	3 18	332	8 43
032	0.81	.130	3.30	.340	8.64
040	1 02	.135	3.43	.400	10 16
045	1.14				

NOTES:
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Unless otherwise specified, tolerance is ±.005 (0 13 mm) on decimals and ±2° on angles
4. These connectors mate with connectors specified in MIL-C-55302/23
5. Use printed circuit boards with the type III receptacle
6. Minimum pad length on printed circuit board
7. Termination layout on .025 (0 64 mm) modular grid
(F) 8. Numbers indicating, every 5 cavities marked or molded on side (except 10 position). Numbers indicating end cavities and lines indicating every 5 cavities molded or marked on mating face
9. Location indicators embossed on surface

(F) FIGURE 1. Connectors, receptacle, 075 (1 90 mm) spacing - Continued

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REQUIREMENTS

Design and construction

Dimensions and configuration See figure 1

Material:

(F) Insulator body: Insulator material shall be in accordance with MIL-C-55302 orMIL-M-24519 type GLCP-30F

(F) Contact material: Contact material shall be in accordance with MIL-C-55302 or beryllium copper alloy CDA174 in accordance with ASTM B768

Plating

Engagement area Gold per MIL-C-55302 for 100 length, over nickel all over per MIL-C-55302

Termination area Tin lead over nickel per MIL-C-55302 for the following lengths

165 minimum length for type I, 200 minimum length for type II
300 minimum length for type III, 140 minimum length for type IV

Contact identification See figure 1

Contact rating 3 0 amperes maximum per contact, 2 25 amperes continuous per contact at 75°F

Keying See MIL-C-55302/31

Oversized pin exclusion (sockets only) 0465 diameter pin

Contact separation force. The connector shall meet the requirements of MIL-C-55302/29

Wire size Number 26 AWG maximum with type II, solder eyelet contact for wire termination,

Resistance to test probe damage The connector shall meet the requirements of MIL-C-55302/30

Mating and unmating The maximum mating force in pounds shall be the number of contacts multiplied by 25 and the withdrawal force in pounds shall be a minimum of 025 times the number of contacts and shall not exceed the measured insertion force

Contact resistance: The average contact resistance of all contacts measured shall not exceed 015 ohm, and no individual contact pair shall have a resistance exceeding 020 ohm

Contact retention: 3 pounds minimum

Dielectric withstanding voltage

Sea level 900 volts rms

High altitude 200 volts rms

Replacement contacts See MIL-C-55302/32

Contact removal tool MIL-I-81969/9-01

Part number M55302/24- (dash number from figure 1)

Patent number 3,404,367 The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government

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CONCLUDING MATERIAL

Custodians
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 99
DLA - ES

User activities
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity
Army - CR

Agent
DLA - ES

(Project 5935-3858-01)

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